

Web-based Pet Care System (PawLife)

Group Members & Scrum Role Assignments

Student ID	Member Name	Assigned Scrum Role	Project Module / Area of Responsibility
IT25100618	Sanvidu S.D.N	Product Owner	System Administration & Business Intelligence Module
IT25101709	De Silva L.P.B	Scrum Master	Digital Pet Profile & Health Record Module
IT25102521	Subasinghe R. A. G. I	Developer 1	Appointment Scheduling & Notification Module
IT25103607	Balasooriya B.A.H.N	Developer 2	Smart Inventory & Medical Supply Chain Management Module
IT25101534	Alahakoon A.M.B.U	Developer 3	Operational Centre & Staff Management Module
IT25103408	Warnakulasuriya G. A. D. T. S	Developer 4	Pet Grooming & Specialty Care Service Workflow Module

1. Identified Personas & User Stories

Persona 1: Tharushi Silva (Business Owner / System Admin)

- **User Story 1:** As a Business Owner, I want to view high level revenue and business growth analytics so that I can make informed strategic expansion decisions.
- **User Story 2:** As a System Admin, I want to configure Role-Based Access Control (RBAC) permissions so that users can only access information relevant to their responsibilities.
- **User Story 3:** As a System Admin, I want to maintain system audit logs so that I can track user actions and maintain data security.
- **User Story 4:** As a Business Owner, I want to analyze customer feedback and service popularity trends so that I can improve overall client satisfaction.

Persona 2: Nadeesha Perera (Pet Care Centre Manager)

- **User Story 1:** As a Centre Manager, I want to view a summary dashboard of daily appointments and staff workload so that I can oversee daily clinic operations efficiently.
- **User Story 2:** As a Centre Manager, I want to receive operational alerts for overdue vaccinations and missed appointments so that staff can quickly follow up.
- **User Story 3:** As a Centre Manager, I want to track staff attendance and shift allocations so that I can ensure optimal resource management.
- **User Story 4:** As a Centre Manager, I want to monitor ongoing service statuses so that clinic bottlenecks are minimized.

Persona 3: Kasun Jayawardena (Veterinary Officer)

- **User Story 1:** As a Veterinary Officer, I want to search and access complete pet health profiles so that I can review past diagnoses, allergies, and medical history.
- **User Story 2:** As a Veterinary Officer, I want to log consultation notes and treatment details so that accurate clinical records are maintained.
- **User Story 3:** As a Veterinary Officer, I want to record vaccination logs and auto-calculate next due dates so that pets receive timely preventative care.
- **User Story 4:** As a Veterinary Officer, I want to issue digital downloadable prescriptions so that pet owners can easily view medication instructions.

Persona 4: Inventory Manager & Inventory Assistant (Balasooriya B.A.H.N)

- **User Story 1:** As an Inventory Manager, I want to track vaccine and medicine stock levels with batch numbers and expiry dates so that clinic supplies remain safe and valid.
- **User Story 2:** As an Inventory Assistant, I want to receive automated reorder alerts when supplies hit low thresholds so that clinic operations are never halted.
- **User Story 3:** As an Inventory Manager, I want to track stock usage during grooming and clinical sessions so that inventory deductions are accurate.
- **User Story 4:** As an Inventory Manager, I want to generate waste and expired stock

audit reports so that financial losses are minimized.

Persona 5: Dilani Fernando & Grooming Staff (Warnakulasuriya G.A.D.T.S)

- **User Story 1:** As a Grooming Specialist, I want to customize grooming service packages based on pet breed and coat condition so that tailored care is provided.
- **User Story 2:** As a Pet Owner, I want to view real-time grooming progress updates on my portal so that I know when my pet is ready for pick-up.
- **User Story 3:** As a Grooming Specialist, I want to log observations regarding skin conditions or behavioral notes during sessions so that vets and owners are informed.
- **User Story 4:** As a Groomer, I want to upload before-and-after photos of grooming sessions so that owners can review the quality of service.

Persona 6: Pet Owner (Primary End User)

- **User Story 1:** As a Pet Owner, I want to create and update online profiles for my pets so that the clinic has accurate information.
- **User Story 2:** As a Pet Owner, I want to book veterinary and grooming sessions online so that I don't have to call or visit in person.
- **User Story 3:** As a Pet Owner, I want to track my pet's grooming and care status in real-time so that I have complete visibility.
- **User Story 4:** As a Pet Owner, I want to view my pet's medical records and prescriptions online so that I can stay up to date with their health.

2. Product Backlog

PBI ID	User Story	Priority
PBI-01	As a Business Owner, I want to view high level revenue and business growth analytics so that I can make informed strategic expansion decisions.	Medium
PBI-02	As a System Admin, I want to configure Role-Based Access Control (RBAC) permissions so that users can only access information relevant to their responsibilities.	High
PBI-03	As a System Admin, I want to maintain system audit logs so that I can track user actions and maintain data security	Medium
PBI-04	As a Business Owner, I want to analyze customer	Low

PBI ID	User Story	Priority
	feedback and service popularity trends so that I can improve overall client satisfaction.	
PBI-05	As a Centre Manager, I want to view a summary dashboard of daily appointments and staff workload so that I can oversee daily clinic operations efficiently.	High
PBI-06	As a Centre Manager, I want to receive operational alerts for overdue vaccinations and missed appointments so that staff can quickly follow up.	High
PBI-07	As a Centre Manager, I want to track staff attendance and shift allocations so that I can ensure optimal resource management.	Medium
PBI-08	As a Centre Manager, I want to monitor ongoing service statuses so that clinic bottlenecks are minimized.	Medium
PBI-09	As a Veterinary Officer, I want to search and access complete pet health profiles so that I can review past diagnoses, allergies, and medical history.	High
PBI-10	As a Veterinary Officer, I want to log consultation notes and treatment details so that accurate clinical records are maintained.	High
PBI-11	As a Veterinary Officer, I want to record vaccination logs and auto-calculate next due dates so that pets receive timely preventative care.	High

PBI ID	User Story	Priority
PBI-12	As a Veterinary Officer, I want to issue digital downloadable prescriptions so that pet owners can easily view medication instructions.	Medium
PBI-13	As an Inventory Manager, I want to track vaccine and medicine stock levels with batch numbers and expiry dates so that clinic supplies remain safe and valid.	High
PBI-14	As an Inventory Assistant, I want to receive automated reorder alerts when supplies hit low thresholds so that clinic operations are never halted.	Medium
PBI-15	As an Inventory Manager, I want to track stock usage during grooming and clinical sessions so that inventory deductions are accurate.	Medium
PBI-16	As an Inventory Manager, I want to generate waste and expired stock audit reports so that financial losses are minimized.	Low
PBI-17	As a Grooming Specialist, I want to customize grooming service packages based on pet breed and coat condition so that tailored care is provided.	Medium
PBI-18	As a Pet Owner, I want to view real-time grooming progress updates on my	Medium

PBI ID	User Story	Priority
	portal so that I know when my pet is ready for pick-up	
PBI-19	As a Grooming Specialist, I want to log observations regarding skin conditions or behavioural notes during sessions so that vets and owners are informed.	Low
PBI-20	As a Groomer, I want to upload before-and-after photos of grooming sessions so that owners can review the quality of service.	Low
PBI-21	As a Pet Owner, I want to create and update online profiles for my pets so that the clinic has accurate information.	High
PBI-22	As a Pet Owner, I want to book veterinary and grooming sessions online so that I don't have to call or visit in person.	High
PBI-23	As a Pet Owner, I want to track my pet's grooming and care status in real-time so that I have complete visibility.	Medium
PBI-24	As a Pet Owner, I want to view my pet's medical records and prescriptions online so that I can stay up to date with their health.	High

3. Sprint Planning

The 24 items in the Product Backlog are grouped into three sprints according to priority, so that the highest value functionality for PawLife is delivered first: Sprint 1 covers all High priority stories, Sprint 2 covers all Medium priority stories, and Sprint 3 covers the remaining Low priority stories.

Sprint 1 - Week 4 - Week 7 (High Priority)

Sprint ID	User Story	Priority
SP1-01	As a System Admin, I want to configure Role Based Access Control (RBAC) permissions so that users can only access information relevant to their responsibilities.	High
SP1-02	As a Centre Manager, I want to view a summary dashboard of daily appointments and staff workload so that I can oversee daily clinic operations efficiently.	High
SP1-03	As a Centre Manager, I want to receive operational alerts for overdue vaccinations and missed appointments so that staff can quickly follow up.	High
SP1-04	As a Veterinary Officer, I want to search and access complete pet health profiles so that I can review past diagnoses, allergies, and medical history.	High
SP1-05	As a Veterinary Officer, I want to log consultation notes and treatment details so that accurate clinical records are maintained.	High
SP1-06	As a Veterinary Officer, I want to record vaccination logs and auto-calculate next due dates so that pets receive timely preventative care.	High
SP1-07	As an Inventory Manager, I want to track vaccine and medicine stock levels with batch numbers and expiry dates so that clinic supplies remain safe and valid.	High

SP1-08	As a Pet Owner, I want to create and update online profiles for my pets so that the clinic has accurate information.	High
SP1-09	As a Pet Owner, I want to book veterinary and grooming sessions online so that I don't have to call or visit in person.	High
SP1-10	As a Pet Owner, I want to view my pet's medical records and prescriptions online so that I can stay up to date with their health.	High

Sprint 2 - Week 8 - Week 11 (Medium Priority)

Sprint ID	User Story	Priority
SP2-01	As a Business Owner, I want to view high level revenue and business growth analytics so that I can make informed strategic expansion decisions.	Medium
SP2-02	As a System Admin, I want to maintain system audit logs so that I can track user actions and maintain data security.	Medium
SP2-03	As a Centre Manager, I want to track staff attendance and shift allocations so that I can ensure optimal resource management.	Medium
SP2-04	As a Centre Manager, I want to monitor ongoing service statuses so that clinic bottlenecks are minimized.	Medium
SP2-05	As a Veterinary Officer, I want to issue digital downloadable prescriptions so that pet owners can easily view medication instructions.	Medium
SP2-06	As an Inventory Assistant, I want to receive automated reorder alerts when supplies hit low thresholds so that clinic operations are never halted.	Medium

SP2-07	As an Inventory Manager, I want to track stock usage during grooming and clinical sessions so that inventory deductions are accurate.	Medium
SP2-08	As a Grooming Specialist, I want to customize grooming service packages based on pet breed and coat condition so that tailored care is provided.	Medium
SP2-09	As a Pet Owner, I want to view real-time grooming progress updates on my portal so that I know when my pet is ready for pick-up.	Medium
SP2-10	As a Pet Owner, I want to track my pet's grooming and care status in real-time so that I have complete visibility.	Medium

Sprint 3 - Week 12 - Week 14 (Low Priority)

Sprint ID	User Story	Priority
SP3-01	As a Business Owner, I want to analyze customer feedback and service popularity trends so that I can improve overall client satisfaction.	Low
SP3-02	As an Inventory Manager, I want to generate waste and expired stock audit reports so that financial losses are minimized.	Low
SP3-03	As a Grooming Specialist, I want to log observations regarding skin conditions or behavioral notes during sessions so that vets and owners are informed.	Low
SP3-04	As a Groomer, I want to upload before-and-after photos of grooming sessions so that owners can review the quality of service.	Low

4. Sprint Backlog - Task Breakdown

Each user story selected for a sprint is broken down into concrete development tasks with time estimates (in hours), covering UI design, backend/API development, business logic, integration, and testing.

Sprint 1 Backlog (High Priority)

User Story 1 (PBI-02): As a System Admin, I want to configure Role-Based Access Control (RBAC) permissions so that users can only access information relevant to their responsibilities.

Task ID	Task	Est. Hours
T-01.1	Design role hierarchy and permission matrix (Admin, Manager, Vet, Inventory, Groomer, Pet Owner)	4
T-01.2	Develop backend RBAC engine to enforce role-based route and data access	8
T-01.3	Build admin UI for assigning and editing roles and permissions	6
T-01.4	Implement permission-validation middleware on API endpoints	5
T-01.5	Test RBAC enforcement across all user roles	4
	Total Estimated Hours	27

User Story 2 (PBI-05): As a Centre Manager, I want to view a summary dashboard of daily appointments and staff workload so that I can oversee daily clinic operations efficiently.

Task ID	Task	Est. Hours
T-02.1	Design dashboard UI wireframe (appointments and staff workload widgets)	5

T-02.2	Develop backend API to aggregate daily appointment and staff data	7
T-02.3	Implement real-time data refresh for dashboard widgets	4
T-02.4	Build staff workload visualization (charts and graphs)	5
T-02.5	Test dashboard accuracy with sample appointment data	3
	Total Estimated Hours	24

User Story 3 (PBI-06): As a Centre Manager, I want to receive operational alerts for overdue vaccinations and missed appointments so that staff can quickly follow up.

Task ID	Task	Est. Hours
T-03.1	Design alert rules engine for overdue vaccinations and missed appointments	4
T-03.2	Develop backend scheduled job to scan records and trigger alerts	6
T-03.3	Build alert notification UI (in-app banner and list)	4
T-03.4	Integrate email/SMS alert delivery to Centre Manager	4
T-03.5	Test alert triggering with overdue and missed-appointment scenarios	3
	Total Estimated Hours	21

User Story 4 (PBI-09): As a Veterinary Officer, I want to search and access complete pet health profiles so that I can review past diagnoses, allergies, and medical history.

Task ID	Task	Est. Hours
T-04.1	Design pet health profile search UI with filters	5
T-04.2	Implement backend search API (by pet name, owner, ID)	6
T-04.3	Build detailed profile view (diagnoses, allergies, medical history)	6
T-04.4	Implement access control so only authorized vets view records	3
T-04.5	Test search accuracy and profile data completeness	3
	Total Estimated Hours	23

User Story 5 (PBI-10): As a Veterinary Officer, I want to log consultation notes and treatment details so that accurate clinical records are maintained.

Task ID	Task	Est. Hours
T-05.1	Design consultation note entry form	4
T-05.2	Develop backend API to save and store consultation records	6
T-05.3	Link consultation notes to pet's medical history timeline	4
T-05.4	Implement auto-save/draft feature for long consultations	3
T-05.5	Test consultation logging and record retrieval	3
	Total Estimated Hours	20

User Story 6 (PBI-11): As a Veterinary Officer, I want to record vaccination logs and auto-calculate next due dates so that pets receive timely preventative care.

Task ID	Task	Est. Hours
T-06.1	Design vaccination log entry UI	4
T-06.2	Develop backend API to store vaccination records	5
T-06.3	Implement auto-calculation logic for next due date based on vaccine type	5
T-06.4	Link vaccination due dates to the alert/notification system (PBI-06)	3
T-06.5	Test due-date calculations across multiple vaccine types	3
	Total Estimated Hours	20

User Story 7 (PBI-13): As an Inventory Manager, I want to track vaccine and medicine stock levels with batch numbers and expiry dates so that clinic supplies remain safe and valid.

Task ID	Task	Est. Hours
T-07.1	Design inventory tracking UI (batch number and expiry date fields)	5
T-07.2	Develop backend API to create and update stock records	7
T-07.3	Implement expiry-date validation and batch tracking logic	5
T-07.4	Build low-stock/near-expiry flagging in the UI	4
T-07.5	Test stock tracking with sample batch and expiry data	3
	Total Estimated Hours	24

User Story 8 (PBI-21): As a Pet Owner, I want to create and update online profiles for my pets so that the clinic has accurate information.

Task ID	Task	Est. Hours
T-08.1	Design pet profile creation/edit form (breed, age, medical info)	4
T-08.2	Develop backend API to store and update pet profiles	6
T-08.3	Implement photo upload for pet profile	3
T-08.4	Validate required fields and data formats	2
T-08.5	Test profile creation and update flow	3
	Total Estimated Hours	18

User Story 9 (PBI-22): As a Pet Owner, I want to book veterinary and grooming sessions online so that I don't have to call or visit in person.

Task ID	Task	Est. Hours
T-09.1	Design booking interface (service type, date/time picker)	5
T-09.2	Develop backend API to check staff and slot availability	7
T-09.3	Implement booking confirmation and calendar sync	5
T-09.4	Integrate email/notification confirmation to pet owner	3
T-09.5	Test booking flow for conflicts and availability edge cases	4
	Total Estimated Hours	24

User Story 10 (PBI-24): As a Pet Owner, I want to view my pet's medical records and prescriptions online so that I can stay up to date with their health.

Task ID	Task	Est. Hours
T-10.1	Design medical records and prescriptions viewing UI	4
T-10.2	Develop backend API to fetch owner's pet medical history	5
T-10.3	Implement downloadable/printable prescription view	3
T-10.4	Implement access control so owners see only their pets' records	3
T-10.5	Test record retrieval and access restrictions	3
	Total Estimated Hours	18

Sprint 2 Backlog (Medium Priority)

User Story 1 (PBI-01): As a Business Owner, I want to view high level revenue and business growth analytics so that I can make informed strategic expansion decisions.

Task ID	Task	Est. Hours
T-01.1	Design analytics dashboard UI (revenue trends, growth charts)	5
T-01.2	Develop backend API to aggregate revenue data across modules	8
T-01.3	Implement data visualization for growth metrics	5
T-01.4	Add filters (date range, service type) to the analytics view	3
T-01.5	Test analytics accuracy against sample transaction data	3

	Total Estimated Hours	24
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User Story 2 (PBI-03): As a System Admin, I want to maintain system audit logs so that I can track user actions and maintain data security.

Task ID	Task	Est. Hours
T-02.1	Design audit log data model (user, action, timestamp)	3
T-02.2	Implement backend logging middleware across system modules	6
T-02.3	Build admin UI to view, search, and filter audit logs	5
T-02.4	Implement log export functionality (CSV/PDF)	3
T-02.5	Test log capture accuracy and completeness	3
	Total Estimated Hours	20

User Story 3 (PBI-07): As a Centre Manager, I want to track staff attendance and shift allocations so that I can ensure optimal resource management.

Task ID	Task	Est. Hours
T-03.1	Design staff attendance and shift scheduling UI	5
T-03.2	Develop backend API for attendance check-in/out and shift assignment	6
T-03.3	Implement shift conflict detection logic	4
T-03.4	Build attendance and shift summary reports	3

T-03.5	Test attendance tracking and shift allocation scenarios	3
	Total Estimated Hours	21

User Story 4 (PBI-08): As a Centre Manager, I want to monitor ongoing service statuses so that clinic bottlenecks are minimized.

Task ID	Task	Est. Hours
T-04.1	Design service status tracking UI (in-progress, completed, delayed)	4
T-04.2	Develop backend API to update and query service status in real time	6
T-04.3	Implement bottleneck detection and highlighting logic	4
T-04.4	Test status updates across concurrent services	3
	Total Estimated Hours	17

User Story 5 (PBI-12): As a Veterinary Officer, I want to issue digital downloadable prescriptions so that pet owners can easily view medication instructions.

Task ID	Task	Est. Hours
T-05.1	Design prescription creation form (medication, dosage, instructions)	4
T-05.2	Develop backend API to generate and store prescriptions	5
T-05.3	Implement PDF generation for downloadable prescriptions	5

T-05.4	Link prescriptions to the pet owner's medical record view (PBI-24)	2
T-05.5	Test prescription generation and download	3
	Total Estimated Hours	19

User Story 6 (PBI-14): As an Inventory Assistant, I want to receive automated reorder alerts when supplies hit low thresholds so that clinic operations are never halted.

Task ID	Task	Est. Hours
T-06.1	Define low-stock threshold rules per item type	3
T-06.2	Develop backend scheduled job to check stock levels	5
T-06.3	Implement automated alert generation and delivery	4
T-06.4	Build UI to configure and view reorder thresholds	3
T-06.5	Test alert triggers with varying stock levels	3
	Total Estimated Hours	18

User Story 7 (PBI-15): As an Inventory Manager, I want to track stock usage during grooming and clinical sessions so that inventory deductions are accurate.

Task ID	Task	Est. Hours
T-07.1	Design stock-usage logging UI linked to sessions	4
T-07.2	Develop backend API to deduct inventory upon service use	6

T-07.3	Link stock deductions to grooming and clinical session records	4
T-07.4	Test inventory deduction accuracy across session types	3
	Total Estimated Hours	17

User Story 8 (PBI-17): As a Grooming Specialist, I want to customize grooming service packages based on pet breed and coat condition so that tailored care is provided.

Task ID	Task	Est. Hours
T-08.1	Design package customization UI (breed and coat condition options)	4
T-08.2	Develop backend API to create and store custom service packages	5
T-08.3	Implement pricing calculation based on package selection	3
T-08.4	Test package customization and pricing accuracy	3
	Total Estimated Hours	15

User Story 9 (PBI-18): As a Pet Owner, I want to view real-time grooming progress updates on my portal so that I know when my pet is ready for pick-up.

Task ID	Task	Est. Hours
T-09.1	Design grooming progress tracker UI (check-in, bath, cut, ready)	4
T-09.2	Develop backend API to update and push progress status	6

T-09.3	Implement real-time update mechanism (polling/push notification)	4
T-09.4	Test progress updates from groomer input to owner view	3
	Total Estimated Hours	17

User Story 10 (PBI-23): As a Pet Owner, I want to track my pet's grooming and care status in real-time so that I have complete visibility.

Task ID	Task	Est. Hours
T-10.1	Design combined care/grooming status view for pet owners	4
T-10.2	Develop backend API to consolidate grooming and clinical status	5
T-10.3	Implement real-time sync between staff updates and the owner portal	4
T-10.4	Test end-to-end status visibility for pet owners	3
	Total Estimated Hours	16

Sprint 3 Backlog (Low Priority)

User Story 1 (PBI-04): As a Business Owner, I want to analyze customer feedback and service popularity trends so that I can improve overall client satisfaction.

Task ID	Task	Est. Hours
T-01.1	Design feedback collection form (rating, comments)	3

T-01.2	Develop backend API to store and aggregate feedback data	5
T-01.3	Implement service popularity trend analysis and visualization	5
T-01.4	Test feedback analytics with sample review data	3
	Total Estimated Hours	16

User Story 2 (PBI-16): As an Inventory Manager, I want to generate waste and expired stock audit reports so that financial losses are minimized.

Task ID	Task	Est. Hours
T-02.1	Design audit report template (waste and expired stock)	3
T-02.2	Develop backend API to identify expired and wasted stock entries	5
T-02.3	Implement report generation (PDF/Excel export)	4
T-02.4	Test report accuracy against sample inventory data	3
	Total Estimated Hours	15

User Story 3 (PBI-19): As a Grooming Specialist, I want to log observations regarding skin conditions or behavioral notes during sessions so that vets and owners are informed.

Task ID	Task	Est. Hours
T-03.1	Design observation notes entry UI during grooming sessions	3
T-03.2	Develop backend API to save observation notes to the pet profile	4

T-03.3	Link observations to the veterinary review workflow	3
T-03.4	Test observation logging and visibility to vets and owners	2
	Total Estimated Hours	12

User Story 4 (PBI-20): As a Groomer, I want to upload before-and-after photos of grooming sessions so that owners can review the quality of service.

Task ID	Task	Est. Hours
T-04.1	Design photo upload UI (before-and-after pairs)	3
T-04.2	Develop backend API to store and associate photos with sessions	5
T-04.3	Implement image compression and optimization on upload	3
T-04.4	Test photo upload and display in the owner portal	2
	Total Estimated Hours	13